

A Reflexive Root-Polytope Model for Preorder h -Polynomials

Simplicial-polytopal realization, palindromicity, and unimodality

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Abstract

For a finite preorder τ on $[n]$, let $h_\tau(t)$ count the integer points of its preorder polytope by the number of nonzero coordinates. We give a proposed general proof that this polynomial is the h -polynomial of an n -dimensional simplicial polytope, establishing Conjecture 5.1(c), and hence (a) and (b), of Athanasiadis and Chapoton. The principal construction is the explicit n -dimensional lattice polytope

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : 1 \leq i \leq n\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}).$$

We prove $h^*(C_\tau, t) = h_\tau(t)$ by an integer-fiber decomposition of an augmented bipartite root polytope. Every boundary pulling triangulation of C_τ , coned from the origin, is unimodular; its boundary is combinatorially polytopal. We also prove reflexivity, an exact formula for the gauge of C_τ , and a Gorenstein translation formula for the larger root polytope. The higher-dimensional root-polytope identity and preorder duality are already results of Dai, Hou, Liu, Thawinrak, and Wang, and are not claimed as new here. A separate hypertree argument reconstructs that known identity. Exact verification code and complete small-case data accompany the manuscript.

Scope and status. The intended new result is the general simplicial-polytopal realization, not the already established duality theorem. The proof uses explicitly identified results on support-enumerators and pulling triangulations. All steps specific to the construction are proved below; the external results are identified explicitly. The manuscript has not been independently refereed or checked by a proof assistant. The finite computations test the constructions; they do not replace the proof. Flag realizability, γ -positivity, and real-rootedness are not proved.

Contents

1	The problem and the result	3
2	The augmented bipartite root polytope	4
3	A complete inequality description	5
4	The dimension-reducing lattice fibers	6
5	Reflexivity and simplicial-polytopal realization	8
6	A second geometric check in dimension $2n$	9
7	Consequences and quantitative formulas	10
8	Examples and the role of transitivity	12
9	Exact computational verification	13
10	What remains unresolved and how to audit the claim	14
A	A hypertree proof of the known root identity	15
B	Proof dependencies and normalization audit	16
C	Reproducing the accompanying computations	17

Reading route. Sections 1–5 contain the main argument. Section 6 gives a second geometric check. Sections 8–10 provide examples, exact experiments, and limitations. Appendix A gives an alternative proof of the known higher-dimensional identity; Appendix B records the proof dependencies and the main normalization checks.

1 The problem and the result

A *preorder* is a reflexive, transitive binary relation; antisymmetry is not required. For a preorder τ on $E = [n]$, write $\mathcal{J}(\tau)$ for its order ideals, including $\emptyset$ and E . For a vector x and a subset $A \subseteq E$, write $x(A) = \sum_{i \in A} x_i$. The preorder polytope and its support polynomial are

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}_{\geq 0}^n : x(I) \leq |I| \text{ for every } I \in \mathcal{J}(\tau)\}, \quad (1.1)$$

$$P_\tau = \mathcal{Q}_\tau \cap \mathbb{Z}^n, \quad h_\tau(t) = \sum_{x \in P_\tau} t^{|\text{supp}(x)|} = \sum_{k=0}^n h_k(\tau) t^k. \quad (1.1)$$

Here $\text{supp}(x) = \{i : x_i \neq 0\}$. These are the objects of Athanasiadis and Chapoton [1]. Their Conjecture 5.1(c) asks whether $(h_0(\tau), \dots, h_n(\tau))$ is the h -vector of an n -dimensional simplicial polytope. Parts (a) and (b) ask, respectively, for palindromicity and palindromic unimodality. The stronger part (d) requires a flag simplicial polytope.

1.1 What was already known when this problem was selected

The arXiv version of [1] is dated 26 May 2026. A subsequent paper by Dai, Hou, Liu, Thawinrak, and Wang [2], in its 27 August 2026 revision, proves that the support-enumerator of a bipartite graph is the Ehrhart numerator of its augmented root polytope. Its Theorem 1.3 explicitly settles Conjecture 5.4 of [1], the identity $h_\tau = h_{\tau^*}$ under reversal of the preorder. These facts must therefore be treated as prior results, not as open problems. Problem 5.3 of [2] still discusses the coefficient-shape conjectures for preorders. Menon's August 2026 paper [6] gives further special cases of γ -positivity for arbors.

The target of the present manuscript is the *general* assertion in Conjecture 5.1(c). The advance proposed here is a dimension-reducing lattice fiber decomposition and the resulting explicit root-polytope realization. The literature check establishes these version-specific statements; it does not certify absolute priority against every unpublished or unindexed argument.

Theorem 1.1 (Main theorem). *Let τ be any preorder on $[n]$, where $n \geq 1$, and put*

$$C_\tau = \text{conv}(\{\pm \mathbf{e}_i : i \in [n]\} \cup \{\mathbf{e}_i - \mathbf{e}_j : j \leq_\tau i, i \neq j\}) \subset \mathbb{R}^n. \quad (1.2)$$

Then C_τ is reflexive and

$$h^*(C_\tau, t) = h_\tau(t). \quad (1.3)$$

Moreover, any pulling triangulation of ∂C_τ is combinatorially the boundary of an n -dimensional simplicial polytope S_τ satisfying

$$h(S_\tau, t) = h_\tau(t).$$

Consequently $h_k(\tau) = h_{n-k}(\tau)$ for all k , and

$$1 = h_0(\tau) \leq h_1(\tau) \leq \dots \leq h_{\lfloor n/2 \rfloor}(\tau).$$

In fact the full g -theorem conditions hold for $h(\tau)$.

A pulling triangulation is obtained by fixing an order on the vertices and recursively triangulating faces by cones from their first vertex. The actual root polytope C_τ need not be simplicial. The theorem asserts that its triangulated boundary has a simplicial convex realization with the desired h -vector. These two assertions should not be conflated.

The empty preorder is harmless: its support polynomial is 1, and both root models degenerate to a point. We treat $n \geq 1$ below to avoid unnecessary zero-dimensional boundary conventions.

1.2 A few useful sanity checks

Since E is an ideal, every $x \in P_\tau$ has $x(E) \leq n$. Hence P_τ is finite, $h_0 = 1$, and $h_n = 1$: the only point with all coordinates positive is $\mathbf{1}$. For a point supported just at i , the largest allowed value of its coordinate is $|\downarrow i|$, where $\downarrow i = \{j : j \leq_\tau i\}$. Therefore

$$h_1(\tau) = \sum_{i=1}^n |\downarrow i| = |\{(j, i) : j \leq_\tau i\}|. \quad (1.4)$$

Elements in the same equivalence class of the preorder remain separate coordinates throughout. Replacing a preorder by its quotient poset without retaining class sizes would change the problem.

2 The augmented bipartite root polytope

Let G_τ have left vertices $1_L, \dots, n_L$, right vertices $1_R, \dots, n_R$, and an edge $i_L j_R$ precisely when $j \leq_\tau i$. Adjoin 0_L and 0_R , with 0_L adjacent to every right vertex and 0_R adjacent to every left vertex. Denote the resulting graph by $\widehat{G}_\tau$. In particular $0_L 0_R$ is an edge. Its root polytope is

$$B_\tau = \text{conv}\{\mathbf{e}_i + \mathbf{f}_j : i_L j_R \in E(\widehat{G}_\tau)\} \subset \mathbb{R}^{n+1} \oplus \mathbb{R}^{n+1}. \quad (2.1)$$

The two coordinate blocks will be denoted (u, v) . We use the plus-sign convention for bipartite roots. Negating the second block converts this into the difference convention and preserves the lattice.

The affine span of B_τ is

$$H_1 = \{(u, v) : \sum_{i=0}^n u_i = \sum_{j=0}^n v_j = 1\}. \quad (2.2)$$

Indeed, the edges $i_L 0_R$ and $0_L j_R$ form a spanning tree, whose $2n + 1$ root vertices are affinely independent. Thus $\dim B_\tau = 2n$. At dilation m , the lattice is the full integer lattice in $H_m = \{\sum u_i = \sum v_j = m\}$, not a smaller lattice generated by selected coordinates.

2.1 The known support-enumerator theorem

For a bipartite graph with left neighborhoods $D_1, \dots, D_n$ in an n -element right shore, its draconian vectors are the nonnegative integer vectors x satisfying

$$x(A) \leq \left| \bigcup_{i \in A} D_i \right| \quad (A \subseteq [n]). \quad (2.3)$$

Theorem 1.2 of [2] identifies their support-enumerator with the h^* -polynomial of the graph augmented by the two universal vertices just used in (2.1).

Lemma 2.1. *For $D_i = \downarrow i$, the draconian vectors in (2.3) are exactly P_τ .*

Proof. If $x \in P_\tau$ and $A \subseteq E$, transitivity makes $\downarrow A$ an ideal. Nonnegativity gives

$$x(A) \leq x(\downarrow A) \leq |\downarrow A| = \left| \bigcup_{i \in A} D_i \right|.$$

Conversely, if (2.3) holds and I is an ideal, reflexivity and downward closure give $\bigcup_{i \in I} D_i = I$. Taking $A = I$ gives $x(I) \leq |I|$. $\square$

Proposition 2.2 (Known higher-dimensional identity). *For every preorder on $[n]$,*

$$\text{Ehr}_{B_\tau}(t) := \sum_{m \geq 0} |mB_\tau \cap \mathbb{Z}^{2n+2}| t^m = \frac{h_\tau(t)}{(1-t)^{2n+1}}. \quad (2.4)$$

Proof. Apply the support-enumerator theorem of [2] and Lemma 2.1. $\square$

Appendix A supplies another proof of this proposition using the Kálmán–Postnikov theorem [3]. Its elementary matching construction also explains why a support statistic, rather than a sum-of-coordinates statistic, occurs. Neither proof makes the already known proposition a novelty claim of this manuscript.

3 A complete inequality description

The special feature of a preorder is that all relevant neighborhood sets can be reduced to ideals. We need the reduction over the reals, not only for integer degree vectors.

Lemma 3.1 (Weighted Hall criterion). *Let G be a bipartite graph with shores L, R . Nonnegative margins u and v with common total m are the row and column sums of a nonnegative matrix supported on the edges of G if and only if*

$$u(A) \leq v(N(A)) \quad (A \subseteq L). \quad (3.1)$$

If the margins are integral, the matrix can be chosen integral.

Proof. Necessity follows by summing the entries in the rows of A . For sufficiency, make a network with source-to-left capacities u_i , right-to-sink capacities v_j , and capacity $m + 1$ on each allowed left-to-right edge. A cut of capacity at most m cannot cut an allowed left-to-right edge. If its source side contains the left set A , it must therefore contain $N(A)$ on the right; its capacity is at least $m - u(A) + v(N(A)) \geq m$. A flow of value m follows from the max-flow/min-cut argument. For completeness, that argument chooses a maximum feasible flow, which exists by compactness; if its residual graph contains a source-to-sink path, a positive augmentation contradicts maximality. Otherwise the vertices reachable from the source give a cut with capacity equal to the flow value. Integral capacities allow integral augmentations from the zero flow, so an integral maximum flow exists in the integral case. $\square$

Proposition 3.2 (Ideal inequalities for B_τ). *For every real $m \geq 0$, a pair $(u, v) \in H_m$ belongs to mB_τ exactly when all its coordinates are nonnegative and*

$$u(I) - v(I) \leq v_0 \quad (I \in \mathcal{J}(\tau)). \quad (3.2)$$

Here $u(I), v(I)$ involve only the coordinates indexed by $[n]$.

Proof. A point of mB_τ is precisely the pair of margins of a nonnegative matrix of total m supported on $\widehat{G}_\tau$. For a left subset containing 0_L , its neighborhood is the entire right shore, so (3.1) follows from nonnegativity and the total-sum condition. For $A \subseteq [n]$,

$$N(A) = \{0_R\} \cup (\downarrow A)_R.$$

Thus the remaining Hall conditions are $u(A) \leq v_0 + v(\downarrow A)$. If (3.2) holds, then

$$u(A) \leq u(\downarrow A) \leq v_0 + v(\downarrow A),$$

so all Hall conditions hold. Conversely, taking $A = I$ for an ideal gives $\downarrow I = I$ and recovers (3.2). The case $m = 0$ is immediate. $\square$

The empty-ideal inequality is just $v_0 \geq 0$, and the full-ideal inequality is equivalent to $u_0 \geq 0$. They may be redundant, but retaining them makes later formulas uniform. We do not assert that every inequality in this presentation defines a distinct facet.

4 The dimension-reducing lattice fibers

Define the projection

$$\pi(u, v) = (u_i - v_i)_{i=1}^n. \quad (4.1)$$

On the root vertices in (2.1), it has values 0, $\pm \mathbf{e}_i$, and $\mathbf{e}_i - \mathbf{e}_j$ for $j \leq_\tau i$. Therefore

$$\pi(B_\tau) = C_\tau. \quad (4.2)$$

Its kernel on the direction space of H_1 is spanned by $\mathbf{e}_i + \mathbf{f}_i - \mathbf{e}_0 - \mathbf{f}_0$, $1 \leq i \leq n$. The dimension loss is exactly n . A linear projection alone does *not* usually preserve an Ehrhart numerator. The following explicit fibers are what make this one work.

For $y \in \mathbb{R}^n$, put

$$\begin{aligned} y_i^+ &= \max(y_i, 0), & y_i^- &= \max(-y_i, 0), \\ p(y) &= \sum_i y_i^+, & q(y) &= \sum_i y_i^-, \end{aligned} \quad (4.3)$$

$$a_\tau(y) = \max_{I \in \mathcal{J}(\tau)} y(I), \quad g_\tau(y) = q(y) + a_\tau(y). \quad (4.4)$$

The empty and full ideals imply $a_\tau(y) \geq 0$ and $a_\tau(y) \geq p(y) - q(y)$.

Theorem 4.1 (Exact fibers and an integral gauge). *For $y \in \mathbb{R}^n$, define*

$$\begin{aligned} u^{\min}(y) &= (a_\tau(y) + q(y) - p(y), y_1^+, \dots, y_n^+), \\ v^{\min}(y) &= (a_\tau(y), y_1^-, \dots, y_n^-). \end{aligned} \quad (4.5)$$

Every point of mB_τ projecting to y has a unique expression

$$(u, v) = (u^{\min}(y), v^{\min}(y)) + \sum_{i=0}^n w_i(\mathbf{e}_i + \mathbf{f}_i), \quad (4.6)$$

where $w_i \geq 0$ and

$$\sum_{i=0}^n w_i = m - g_\tau(y). \quad (4.7)$$

Consequently,

$$y \in mC_\tau \iff g_\tau(y) \leq m. \quad (4.8)$$

For integral y, m , all terms in (4.5) are integral, and the lattice points in the fiber correspond exactly to nonnegative integer vectors w satisfying (4.7).

Proof. The two vectors in (4.5) are nonnegative. Their sums are both $a_\tau(y) + q(y) = g_\tau(y)$, and their difference on $[n]$ is y . Moreover,

$$u^{\min}(I) - v^{\min}(I) = y(I) \leq a_\tau(y) = v_0^{\min}.$$

Proposition 3.2 puts this pair in $g_\tau(y)B_\tau$. Adding a nonnegative multiple of a diagonal root leaves every difference $u(I) - v(I)$ unchanged. For the root indexed by 0, the right side v_0 also

increases, so all inequalities remain valid. This proves that every expression in (4.6) with (4.7) is an admissible point.

Conversely, suppose $(u, v) \in mB_\tau$ projects to y . For $1 \leq i \leq n$, nonnegativity and $u_i - v_i = y_i$ imply uniquely

$$u_i = y_i^+ + w_i, \quad v_i = y_i^- + w_i, \quad w_i = \min(u_i, v_i) \geq 0.$$

The ideal inequalities give $v_0 \geq \max_I y(I) = a_\tau(y)$. Set $w_0 = v_0 - a_\tau(y) \geq 0$. The equality of the two total sums then forces

$$u_0 = v_0 + q(y) - p(y) = u_0^{\min}(y) + w_0.$$

All coordinates in (4.6) are now determined. Summing either block yields (4.7). Projection proves (4.8). If y is integral, the maximum defining $a_\tau(y)$ is an integer, which establishes the final assertion without any lattice-saturation assumption. $\square$

Corollary 4.2 (Equality of Ehrhart numerators). *For every finite preorder,*

$$\text{Ehr}_{B_\tau}(t) = \frac{\text{Ehr}_{C_\tau}(t)}{(1-t)^n}, \quad h^*(B_\tau, t) = h^*(C_\tau, t) = h_\tau(t). \quad (4.9)$$

Proof. For integral y , the least admissible integral dilation is $g_\tau(y)$. Theorem 4.1 gives

$$\text{Ehr}_{B_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{(1-t)^{n+1}}, \quad \text{Ehr}_{C_\tau}(t) = \frac{\sum_{y \in \mathbb{Z}^n} t^{g_\tau(y)}}{1-t}.$$

These are legitimate formal power series: each coefficient receives only finitely many contributions because C_τ is bounded. Dividing the two identities proves the first equality in (4.9). The dimensions are $2n$ and n , respectively, so their Ehrhart numerators coincide. Finally apply Proposition 2.2. $\square$

There is a useful coefficient-level form. The fiber over an integral y at dilation m contains

$$\binom{m - g_\tau(y) + n}{n} \quad \text{if } g_\tau(y) \leq m, \quad 0 \quad \text{otherwise.} \quad (4.10)$$

Thus the factor $(1-t)^{-n}$ is not a heuristic cancellation of dimensions; it comes from a concrete, uniquely parametrized family of integer fibers.

4.1 An order-theoretic interpretation of the gauge

The gauge can also be written

$$g_\tau(y) = \max_{I \in \mathcal{J}(\tau)} \left(\sum_{i \in I} y_i^+ + \sum_{i \notin I} y_i^- \right). \quad (4.11)$$

Thus testing membership in a dilation of C_τ is a maximum-weight-ideal problem. One need not enumerate ideals in a practical implementation: put source capacity y_i^+ at i , sink capacity y_i^- at i , and sufficiently large capacities on arcs $i \rightarrow j$ for $j \leq_\tau i$. A finite minimum cut has a downward-closed source side I , and its capacity is $p(y) - y(I)$. Hence $a_\tau(y)$, and therefore $g_\tau(y)$, can be obtained by a network-flow calculation. The accompanying small-case verifier deliberately uses exhaustive ideal enumeration instead, keeping the check independent of a flow library.

5 Reflexivity and simplicial-polytopal realization

We now finish the proof of Theorem 1.1. The required geometry is particularly simple because all nonzero vertices of C_τ are type- A roots with one coordinate deleted.

5.1 The polar is integral

Since C_τ contains $\text{conv}\{\pm \mathbf{e}_i\}$, it is full-dimensional and contains 0 in its interior. Its polar is

$$C_\tau^\circ = \{z \in \mathbb{R}^n : -1 \leq z_i \leq 1, \quad z_i - z_j \leq 1 \text{ whenever } j \leq_\tau i\}. \quad (5.1)$$

Lemma 5.1 (Root determinants). *Put $\mathbf{e}_0 = 0 \in \mathbb{Z}^n$. Any n linearly independent vectors of the form $\mathbf{e}_i - \mathbf{e}_j$, with $i, j \in \{0, \dots, n\}$ and $i \neq j$, form a basis of $\mathbb{Z}^n$.*

Proof. Associate the vector $\mathbf{e}_i - \mathbf{e}_j$ to a directed edge from j to i on $\{0, \dots, n\}$. A cycle in the underlying undirected graph would give a linear dependence, with signs determined by the edge directions. Consequently n independent vectors give a forest with n edges on $n + 1$ vertices, hence a spanning tree. The matrix of these vectors is a directed incidence matrix with the row indexed by 0 deleted. Remove a leaf different from 0; its row has one nonzero entry, equal to 1 or -1 . Expanding the determinant along that row and repeating reduces the determinant to 1 or -1 . $\square$

Proposition 5.2. *The polar (5.1) is a lattice polytope. In fact all its vertices have coordinates in $\{-1, 0, 1\}$. Thus C_τ is reflexive.*

Proof. At a vertex of the full-dimensional polar, choose n linearly independent active constraints. With $z_0 = 0$, each selected equality is of the form $z_i - z_j = 1$. Its coefficient row is a root as in Lemma 5.1. The resulting square matrix has determinant ± 1 , so its solution is integral. The bounds $-1 \leq z_i \leq 1$ restrict these integers to $-1, 0, 1$. Both the original polytope and its polar are therefore integral, with the origin in the interior, which is precisely reflexivity. $\square$

This proof also supplies a finite exact way to test lattice-point membership: one may evaluate against all ternary vectors satisfying (5.1), since that set contains every vertex of the polar. The verifier uses this method as a check independent of the gauge formula.

5.2 Unimodular cones over the boundary

Recall that a lattice simplex with vertices $0, r_1, \dots, r_n$ is unimodular when $r_1, \dots, r_n$ form a lattice basis. A triangulation is unimodular if all its full-dimensional simplices are unimodular.

Proposition 5.3. *Let Δ be any triangulation of ∂C_τ using only vertices of C_τ . Then the star triangulation $0 * \Delta$ of C_τ is unimodular.*

Proof. Every boundary simplex is contained in a supporting hyperplane not containing 0. The n vertices of a maximal boundary simplex are therefore linearly independent: an affine dependence is already excluded by being a simplex, and a linear dependence with nonzero coefficient sum would place 0 in that supporting hyperplane. All its vertices are roots $\mathbf{e}_i - \mathbf{e}_j$, allowing index 0. Lemma 5.1 shows that the cone simplex from 0 is unimodular. Since 0 is interior, these cones form a triangulation of all of C_τ . $\square$

5.3 Why the triangulated boundary is polytopal

We use the following standard pulling fact, stated as Lemma 2.1 in Athanasiadis [4]: a pulling (reverse lexicographic) triangulation of the boundary of a convex n -polytope is combinatorially the boundary of a simplicial convex n -polytope. The geometric mechanism is successive small outward perturbations of the ordered vertices; at each step one chooses the perturbation small enough not to change the already fixed incidences. The resulting convex boundary realizes the prescribed pulling subdivision. This is an actual polytopality assertion, not merely shellability or the assertion that the complex is a sphere.

Here is the Ehrhart connection, with its normalization made explicit. If K is a d -dimensional unimodular triangulation of a lattice polytope P , then

$$\text{Ehr}_P(t) = \sum_{\sigma \in K} \left(\frac{t}{1-t} \right)^{|\sigma|} = \frac{h(K, t)}{(1-t)^{d+1}}, \quad (5.2)$$

where the empty face contributes 1. Indeed, the relative interior of an r -dimensional unimodular simplex contributes $\binom{m-1}{r}$ lattice points at positive dilation m , with generating series $t^{r+1}/(1-t)^{r+1}$. Relative interiors partition the triangulation, giving the first equality. The second equality is the definition of its h -polynomial. For a cone, the face generating function gains a factor $1 + t/(1-t)$, while the denominator exponent increases by one; hence

$$h(0 * \Delta, t) = h(\Delta, t). \quad (5.3)$$

Completion of the proof of Theorem 1.1. Choose a pulling triangulation Δ of ∂C_τ . By the pulling fact, Δ is the boundary of an n -dimensional simplicial polytope S_τ . Proposition 5.3 and (5.2)–(5.3) give

$$h(S_\tau, t) = h(\Delta, t) = h^*(C_\tau, t) = h_\tau(t),$$

where the last equality is Corollary 4.2. Reflexivity was proved in Proposition 5.2. The Dehn–Sommerville relations for simplicial polytopes give palindromicity, and the g -theorem gives the stated inequalities and its stronger numerical conditions; see also the polytopal conclusions summarized in [4, Theorem 3.5]. $\square$

6 A second geometric check in dimension $2n$

The larger polytope B_τ supplies two independent checks on the preceding argument: an interior-point translation and an explicit special simplex. They are useful safeguards against a sign error or a mismatch of dimensions.

6.1 An exact interior translation

Let $\mathbf{c} = (1, \dots, 1; 1, \dots, 1) \in \mathbb{Z}^{2n+2}$. For every $m \geq n+1$,

$$(\text{relint}(mB_\tau)) \cap \mathbb{Z}^{2n+2} = \mathbf{c} + ((m-n-1)B_\tau \cap \mathbb{Z}^{2n+2}). \quad (6.1)$$

To see this, note that every coordinate inequality is nonconstant on B_τ , and each ideal inequality has strict slack at $\mathbf{e}_0 + \mathbf{f}_0$. Consequently relative interior is characterized by positive coordinates and strict versions of (3.2). At integer points these strict inequalities are

$$u(I) - v(I) \leq v_0 - 1.$$

Subtracting 1 from every coordinate reduces each block sum by $n + 1$. The terms $|I|$ cancel on the left, and the right side becomes the new v_0 . This is exactly the weak inequality description of $(m - n - 1)B_\tau$. Conversely, adding $\mathbf{c}$ makes every defining inequality strict. There are no interior integer points for $1 \leq m < n + 1$, since $n + 1$ positive coordinates in either block already require total at least $n + 1$.

In particular, B_τ has codegree $n + 1$, with unique interior integer point $\mathbf{c}$ in its first nonempty interior dilation. It is Gorenstein of index $n + 1$. Ehrhart reciprocity translates (6.1) into

$$t^{2n+1}h^*(B_\tau, t^{-1}) = t^{n+1}h^*(B_\tau, t),$$

which again gives $h_\tau(t) = t^n h_\tau(t^{-1})$. This symmetry proof does not use the g -theorem or polytopality of a sphere.

6.2 The diagonal special simplex

A simplex with r vertices inside a polytope is *special* if each facet contains exactly $r - 1$ of those vertices. Consider

$$\Sigma_\tau = \text{conv}\{d_i : 0 \leq i \leq n\}, \quad d_i = \mathbf{e}_i + \mathbf{f}_i. \quad (6.2)$$

Reflexivity of the preorder makes these root vertices available, and they are affinely independent. A coordinate hyperplane $u_i = 0$ or $v_i = 0$ contains exactly the diagonal vertices other than d_i . On an ideal hyperplane $u(I) - v(I) = v_0$, every d_i with $i > 0$ lies on the hyperplane and d_0 does not. Every facet is furnished by at least one nonredundant inequality of Proposition 3.2; thus Σ_τ is special.

Every triangulation of a bipartite root polytope using its vertices is unimodular. Indeed, affine dependence of edge roots is equivalent to a cycle in the supporting bipartite graph; a full-dimensional simplex therefore corresponds to a spanning tree. Leaf elimination in its incidence matrix shows that its edge vectors form a basis of the integer lattice $\{(u, v) : \sum u_i = \sum v_j\}$. Intersecting with the height-one hyperplane shows affine unimodularity. In particular every pulling ordering is compressed in the terminology of [4].

Athanasiadis's special-simplex theorem [4, Theorem 3.5], applied with dimension $2n$ and $n + 1$ special vertices, therefore realizes $h^*(B_\tau, t)$ as the h -polynomial of a simplicial polytope of dimension $2n - (n + 1) + 1 = n$. This recovers the main theorem by another route. The broader Gorenstein/unimodular-triangulation framework of Bruns and Römer [5] provides a further compatible interpretation, but is not needed for our main proof.

7 Consequences and quantitative formulas

7.1 Duality, with its prior attribution

Swapping the coordinate blocks identifies B_τ with B_{τ^*} . This is the already established duality mechanism of [2]. In the smaller model it becomes the especially transparent equality

$$C_{\tau^*} = -C_\tau. \quad (7.1)$$

Thus $h_{\tau^*} = h_\tau$. The gauge gives a third check. Ideals of τ^* are complements of ideals of τ , so

$$a_{\tau^*}(-y) = a_\tau(y) - y(E), \quad g_{\tau^*}(-y) = g_\tau(y).$$

This rederivation is included to connect the new lower-dimensional model with the known theorem, not to reclassify Conjecture 5.4 as open.

7.2 Ehrhart polynomials and volumes

The ordinary Ehrhart polynomial of C_τ is now explicit in terms of the preorder support counts:

$$|mC_\tau \cap \mathbb{Z}^n| = \sum_{k=0}^n h_k(\tau) \binom{m+n-k}{n}, \quad m \in \mathbb{Z}_{\geq 0}. \quad (7.2)$$

Similarly,

$$|mB_\tau \cap \mathbb{Z}^{2n+2}| = \sum_{k=0}^n h_k(\tau) \binom{m+2n-k}{2n}. \quad (7.3)$$

A binomial term with nonnegative upper argument smaller than its lower argument is zero. These formulas also extend as polynomial identities in m . Normalized volume, meaning dimension factorial times volume in the relevant lattice, satisfies

$$\text{Vol}_{\mathbb{Z}}(C_\tau) = \text{Vol}_{\mathbb{Z}}(B_\tau) = h_\tau(1) = |P_\tau|. \quad (7.4)$$

The two volumes in this equality live in different dimensions and are both *normalized* volumes.

Remark 5.6 of [1] considers $h_\tau(t)/(1-t)^{n+1}$. Equation (7.2) identifies its coefficients as the Ehrhart polynomial of C_τ . This gives that series a concrete geometric interpretation. It does not prove that its polynomial has all complex roots on a particular vertical line; no such root-location assertion follows just from reflexivity.

7.3 Vertices, elementary bounds, and the g -vector

Every vector displayed in (1.2) is a vertex. To expose $\mathbf{e}_i$, choose a linear functional taking value 1 at coordinate i and a fixed $\varepsilon \in (0, 1)$ at all other coordinates. The negative case is analogous. To expose $\mathbf{e}_i - \mathbf{e}_j$, give coordinates i, j the values 1, -1 and all others the value 0; this root is the unique displayed vector with value 2. Thus

$$f_0(C_\tau) = n + |\{(j, i) : j \leq_\tau i\}| = n + h_1(\tau). \quad (7.5)$$

Pulling adds no vertices, so the same number occurs in S_τ .

The inclusions

$$[0, 1]^n \subseteq Q_\tau \subseteq \{x \geq 0 : x(E) \leq n\}$$

give the useful coefficientwise bounds

$$\binom{n}{k} \leq h_k(\tau) \leq \binom{n}{k}^2 \quad (0 \leq k \leq n). \quad (7.6)$$

For the upper bound, choose the support in $\binom{n}{k}$ ways; the positive integer values on it with total at most n can be chosen in $\binom{n}{k}$ ways by stars and bars. The lower bound uses the corresponding 0–1 points.

For completeness, the full numerical strengthening supplied by polytopality can be stated as follows. Put $g_0 = 1$ and $g_i = h_i - h_{i-1}$ for $1 \leq i \leq \lfloor n/2 \rfloor$. The g -theorem says this sequence is an M -sequence. In particular $g_i \geq 0$ and $g_{i+1} \leq g_i^{\langle i \rangle}$ whenever both sides are relevant. Here, if

$$a = \binom{b_i}{i} + \binom{b_{i-1}}{i-1} + \cdots + \binom{b_j}{j}, \quad b_i > b_{i-1} > \cdots > b_j \geq j \geq 1,$$

is the Macaulay expansion, then $a^{\langle i \rangle} = \binom{b_{i+1}}{i+1} + \cdots + \binom{b_{j+1}}{j+1}$, with $0^{\langle i \rangle} = 0$. These inequalities are stronger than unimodality and follow here from an actual simplicial-polytope realization.

8 Examples and the role of transitivity

8.1 Antichains and a single equivalence class

For an antichain, $Q_\tau = [0, 1]^n$ and

$$h_\tau(t) = (1 + t)^n, \quad C_\tau = \text{conv}\{\pm \mathbf{e}_1, \dots, \pm \mathbf{e}_n\}.$$

The gauge is $\sum_i |y_i|$. The smaller model is already the simplicial cross polytope, so no boundary refinement is needed.

For the indiscrete preorder, in which every two elements are equivalent, $Q_\tau = \{x \geq 0 : x(E) \leq n\}$. The same stars-and-bars argument used above shows

$$h_\tau(t) = \sum_{k=0}^n \binom{n}{k}^2 t^k.$$

Here $B_\tau = \Delta_n \times \Delta_n$, and the lower-dimensional polytope contains all roots $\pm \mathbf{e}_i$ and $\mathbf{e}_i - \mathbf{e}_j$. Its gauge is $\max(p(y), q(y))$, since its only ideals are $\emptyset, E$. These two extreme examples check both the coefficient bounds and the change of Ehrhart denominator.

8.2 A non-self-dual four-element example

Let τ have one minimum below three incomparable maximal elements. Its polynomial, obtained directly from (1.1), is

$$h_\tau(t) = 1 + 7t + 12t^2 + 7t^3 + t^4.$$

For example, for supports of size two not containing the minimum, the three pairs of maximal elements each allow the values $(1, 1), (1, 2), (2, 1)$, giving nine points; the three supports containing the minimum each allow only $(1, 1)$, giving three more. The dual order is not isomorphic to the original, but its directed root polytope is the negative of the original one and its support polynomial agrees. This is a useful test that duality is not merely invariance under relabeling.

8.3 A preorder with a nontrivial equivalence class

Use labels a, b, c, d, e , with $a \sim c, a < e, c < e, b < e$, and $b < d$. No other comparisons hold beyond those forced by reflexivity and transitivity. The principal ideals are

$$\{a, c\}, \quad \{b\}, \quad \{a, c\}, \quad \{b, d\}, \quad \{a, b, c, e\}.$$

Direct enumeration gives

$$h_\tau(t) = 1 + 11t + 34t^2 + 34t^3 + 11t^4 + t^5. \tag{8.1}$$

This agrees with the running example of [1]. There are 92 points in P_τ , 22 vertices of B_τ , and 16 vertices of C_τ . The number 92 is also the normalized volume of either root model.

For $x = (0, 0, 1, 2, 2)$, the matching construction in Appendix A can assign the five demands to the five right vertices as follows: the demand at c uses c_R , the two demands at d use b_R, d_R , and the two demands at e use a_R, e_R . Attach every left vertex to 0_R , then attach each right vertex to its assigned left vertex. This gives the spanning-tree certificate included in the JSON data.

8.4 Selected coefficient vectors

Preorder	n	$(h_0, \dots, h_n)$	$ P_\tau $
Antichain	4	(1, 4, 6, 4, 1)	16
Chain	4	(1, 10, 20, 10, 1)	42
Single equivalence class	4	(1, 16, 36, 16, 1)	70
One minimum, three maxima	4	(1, 7, 12, 7, 1)	28
Diamond	4	(1, 9, 18, 9, 1)	38
Example (8.1)	5	(1, 11, 34, 34, 11, 1)	92
Six-element crown	6	(1, 12, 48, 75, 48, 12, 1)	197

The crown has minimal elements a, b, c and maximal elements A, B, C , with $a, b < A, b, c < B$, and $c, a < C$. Every row of the table is recomputed by the supplied verifier; no floating-point root finder is involved.

8.5 Why arbitrary reflexive relations are not enough

Transitivity cannot simply be dropped from the fiber theorem. Consider the neighborhoods

$$D_1 = \{1\}, \quad D_2 = \{1, 2\}, \quad D_3 = \{2, 3\}.$$

They define a reflexive but nontransitive relation. Its draconian inequalities reduce to

$$x_i \geq 0, \quad x_1 \leq 1, \quad x_1 + x_2 \leq 2, \quad x_3 \leq 2, \quad x_1 + x_2 + x_3 \leq 3.$$

The support counts are (1, 5, 6, 1): one-coordinate supports contribute $1 + 2 + 2 = 5$, two-coordinate supports contribute $1 + 2 + 3 = 6$, and the full support contributes just (1, 1, 1). This is the nonpalindromic example also recorded in [1, Remark 5.5].

The general support-enumerator/root-polytope theorem remains valid for this graph. What fails is the reduction of all weighted Hall inequalities to matching ideal differences $u(I) - v(I)$. In particular, subtracting a common coordinate from the two margins can damage an inequality involving different row and column subsets. Thus the argument does not inadvertently imply palindromicity for all graph support-enumerators.

9 Exact computational verification

The program code/verify.py uses only Python's standard library and integer arithmetic. It stores a preorder by bitmasks

$$D_i = \{j : j \leq_\tau i\}.$$

It enumerates every reflexive relation on $[n]$ and retains precisely those for which $j \in D_i$ implies $D_j \subseteq D_i$. This is a direct test of transitivity. All weak compositions with total at most n are then checked against the ideal inequalities to obtain h_τ .

Ground-set size	Labeled preorders checked	Result
0	1	Pass
1	1	Pass
2	4	Pass
3	29	Pass
4	355	Pass
5	6,942	Pass
Total	7,332	Pass

For each of these preorders, the code checks duality, palindromicity, unimodality, the endpoint coefficients, the relation-count formula for h_1 , and the special-simplex slack identities. These are finite consistency checks, not the reason the theorem holds for arbitrary n .

Stronger, independent comparisons are performed for all 35 preorders on at most three elements. The program enumerates spanning trees directly, deduplicates their left degree-minus-one vectors, and compares them with $\{(n - |x|, x) : x \in P_\tau\}$. It computes internal inactivity by testing all allowed transfers to smaller coordinates, rather than replacing inactivity by support size in the code. Across these instances it enumerates 18,009 spanning trees and checks 399 explicit matching-tree certificates.

For the same cases it counts sums of edge-root vertices directly at every dilation $0 \leq m \leq 2n$. Those counts are compared both with Hall-margin enumeration and with (7.3). Finite differences recover all $2n + 1$ coefficients of the root-polytope Ehrhart numerator, including the predicted zero coefficients above degree n . Independently, integer points of C_τ are counted using the ternary polar description, compared with the gauge formula, and checked through $m = n$ against (7.2). Full-margin fiber identities and the interior translation are also checked exactly.

Five selected four-element preorders undergo full root Ehrhart checks through $m = 8$, quotient checks through $m = 4$, and direct edge-sum checks through $m = 3$. Further examples include the weighted five-element preorder and the six-element crown. The JSON file records all individual results, and the CSV file contains every preorder and support vector through size five.

The original conjecture paper reports tests through larger sizes for some coefficient properties [1]. The present computations are not offered as an improved size record. Their purpose is to compare independent realizations of the newly proposed construction and detect normalization, orientation, or lattice-index errors.

10 What remains unresolved and how to audit the claim

The proof gives an ordinary simplicial polytope, not a flag one. A flag complex is determined by its edges: every collection of pairwise adjacent vertices must form a face. Neither a pulling triangulation nor a general simplicial polytope automatically has that property. Therefore Conjecture 5.1(d) is not settled by the argument.

Likewise, palindromic unimodality and the g -theorem do not establish γ -positivity or real-rootedness. The manuscript proves neither the nonnegativity of all coefficients in

$$h_\tau(t) = \sum_{j=0}^{\lfloor n/2 \rfloor} \gamma_j t^j (1+t)^{n-2j}$$

nor the assertion that every zero of h_τ is real. Accordingly it does not claim to settle Conjectures 5.2 or 5.3 of [1].

For independent mathematical review, the shortest audit is to verify four links: the identification of the draconian vectors with P_τ ; Proposition 3.2; the unique integer fibers in Theorem 4.1; and the determinant argument for cones over a polytopal pulling boundary. Once these are accepted, the cited support-enumerator theorem and the pulling lemma give the asserted simplicial-polytopal realization. Section 6 is a separate geometric cross-check, not a circular use of the desired symmetry.

The result should be understood as a proposed research proof with reproducible support, not as a claim of community acceptance. The distinction between the known August 2026 result and the additional construction developed here is part of the mathematical claim, not a bibliographic afterthought.

A A hypertree proof of the known root identity

This appendix reconstructs Proposition 2.2 without using the support-enumerator theorem of [2]. It is included for conceptual clarity and as an independent route to the same normalization. Its one substantial imported result is the Kálmán–Postnikov theorem [3].

A.1 Hypertrees and internal inactivity

For a connected bipartite graph with left shore L , a left hypertree is a vector $a \in \mathbb{Z}_{\geq 0}^L$ realized by a spanning tree T through $a_i = \deg_T(i_L) - 1$. Distinct trees realizing the same vector contribute only one hypertree. Order the left shore. A coordinate i is internally inactive when there is a smaller coordinate j such that replacing a by $a - \mathbf{e}_i + \mathbf{e}_j$ gives another hypertree. The interior polynomial is the sum of $t^{\text{number of internally inactive coordinates}}$ over hypertrees.

The Kálmán–Postnikov theorem identifies this interior polynomial with the Ehrhart numerator of the bipartite root polytope. Their paper uses a reversed convention for the triangulation h -polynomial in its Theorem 1.1; the Ehrhart-basis formulation immediately following that theorem, equation (1.2), identifies the coefficients with the usual h^* -numerator used here. Thus there is no additional reversal of t in the statement we apply.

Lemma A.1. *The left hypertrees of $\widehat{G}_\tau$ are exactly*

$$a = (n - x(E), x_1, \dots, x_n), \quad x \in P_\tau. \quad (\text{A.1})$$

Proof. A spanning tree of $\widehat{G}_\tau$ has $2n + 1$ edges, so the sum of its left degree-minus-one vector is $(2n + 1) - (n + 1) = n$. For a nonempty $A \subseteq [n]$, the tree edges incident with A_L form a forest on $A_L \cup N(A)$. Their number is $a(A) + |A|$, and a forest on this nonempty vertex set has at most $|A| + |N(A)| - 1$ edges. Hence

$$a(A) \leq |N(A)| - 1 = |\downarrow A|.$$

Lemma 2.1 now shows that $x = (a_1, \dots, a_n)$ belongs to P_τ , and the sum condition determines a_0 .

Conversely, let $x \in P_\tau$. Replace the left element i by x_i distinct clones, each adjacent to the right set $\downarrow i \subseteq [n]$. A set of clones whose labels lie in A has at most $x(A) \leq |\downarrow A|$ members. Thus Hall's matching condition holds, and all clones can be matched to distinct right vertices. This is also a constructive statement: if an augmenting-path matching leaves a clone unmatched, the alternating paths reachable from it produce a set of clones with fewer neighbors, contradicting Hall's condition.

Build a tree by first taking all edges $i_L 0_R$ for $0 \leq i \leq n$. Each right vertex j_R , $j > 0$, is then attached to the left label of its matched clone, or to 0_L if it is unmatched. The first edges form a star; every subsequent edge attaches a new leaf. Hence the result is a spanning tree. Its left degrees are $x_i + 1$ for $i > 0$ and $n - x(E) + 1$ for $i = 0$, exactly as required. $\square$

Lemma A.2. *Order the left vertices with 0 first. For the hypertree in (A.1), internal inactivity is $|\text{supp}(x)|$.*

Proof. The first coordinate is never internally inactive. A zero coordinate cannot transfer a unit, so it is also not internally inactive. If $x_i > 0$, then $x - \mathbf{e}_i \in P_\tau$ because all the defining inequalities are upper bounds and the coordinates remain nonnegative. By Lemma A.1, the vector obtained by transferring one unit from coordinate i to coordinate 0 is another hypertree. Thus every positive nonzero-index coordinate is internally inactive, and there are no others. $\square$

Summing over (A.1) gives an interior polynomial equal to $h_\tau(t)$. The Kálmán–Postnikov theorem identifies it with $h^*(B_\tau, t)$, proving Proposition 2.2 by this second route. Notice that the argument counts hypertree *vectors*, not spanning trees; counting all realizing trees would introduce incorrect multiplicities.

B Proof dependencies and normalization audit

Claim or input	Where it comes from
$h^*(B_\tau) = h_\tau$	Known: [2, Theorem 1.2], specialized using Lemma 2.1. Alternatively Appendix A plus [3].
Ideal Hall presentation	Proposition 3.2; transitivity is used explicitly.
Integer fiber factorization	Theorem 4.1; all coordinates and inverse parameters are displayed.
$h^*(C_\tau) = h_\tau$	Corollary 4.2; a consequence of the fiber factorization and the known root identity.
Unimodularity	Leaf elimination in Lemma 5.1, followed by Proposition 5.3.
Polytopal pulling boundary	The established pulling lemma [4, Lemma 2.1].
Palindromicity, unimodality	Dehn–Sommerville and the g -theorem applied to the constructed simplicial polytope; an independent reciprocity proof of palindromicity is in Section 6.
Alternative polytopal route	The explicit special simplex and [4, Theorem 3.5].

There are several easy places to make an otherwise plausible argument wrong. The support polynomial of Q_τ is not asserted to be $h^*(Q_\tau)$. It is the numerator of the different polytopes B_τ and C_τ . The larger denominator has exponent $2n + 1$ and the smaller one exponent $n + 1$. Both auxiliary vertices, and the edge between them, are required. The edge orientation convention is $j \leq_\tau i \iff i_L j_R$ is present. The integer-fiber parameters consist of $n + 1$ nonnegative entries; the exponent n in (4.9) results only after dividing by the additional slack factor in the Ehrhart series of C_τ .

No argument substitutes a rational equivalence for an integral one. The fiber base is integral because $a_\tau(y)$ is an integer for integral y . The determinant statements refer to the full relevant lattices. Finally, regularity or shellability alone is not used to claim that a sphere is polytopal: the specific pulling-boundary theorem supplies that step.

C Reproducing the accompanying computations

The archive contains `article.tex`, `article.pdf`, a build script, `code/verify.py`, the full CSV enumeration, a JSON result report, and status notes. From the extracted directory, run

```
python3 code/verify.py --output data
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verifier requires Python 3.10 or later and no third-party Python packages. Do not use Python's `-O` option, since assertions are part of the checks. The LaTeX document uses standard TeX Live packages, including `newpxtext`, `newpxmath`, `tcolorbox`, and `hyperref`.

The CSV uses zero-based bit positions: the row value 5 represents the set $\{0, 2\}$. Row i records the principal ideal of element i , not its filter. A polynomial is stored as coefficients from degree zero upward. The JSON report also records explicit edge sets for matching certificates and complete Ehrhart counts for the independently tested instances.

The calculations in the supplied report completed without failed assertions. Their runtime is recorded separately as an observation from this environment, not as a performance guarantee. No claim of formal verification is made.

References

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